

DATA/CS 172: Python for Data Analysis

If Statements

Prof. Travis Mandel

9/6/2024



UNIVERSITY
of HAWAII®

HILO

Reminders

- Lab 1 final deadline today
- Lab 2 due today
- Please seek out help if you are confused about anything!

Agenda

- Booleans
- Logic
- If statements

Right now...

Our scripts run top to bottom

Every line runs every time

What about **conditionals**/ branches? ← Important term!

If **something** then do thing 1, otherwise do thing 2

Example: Data might have values that don't make sense, if we detect one of these we should handle things differently

But what can we write for **something**?

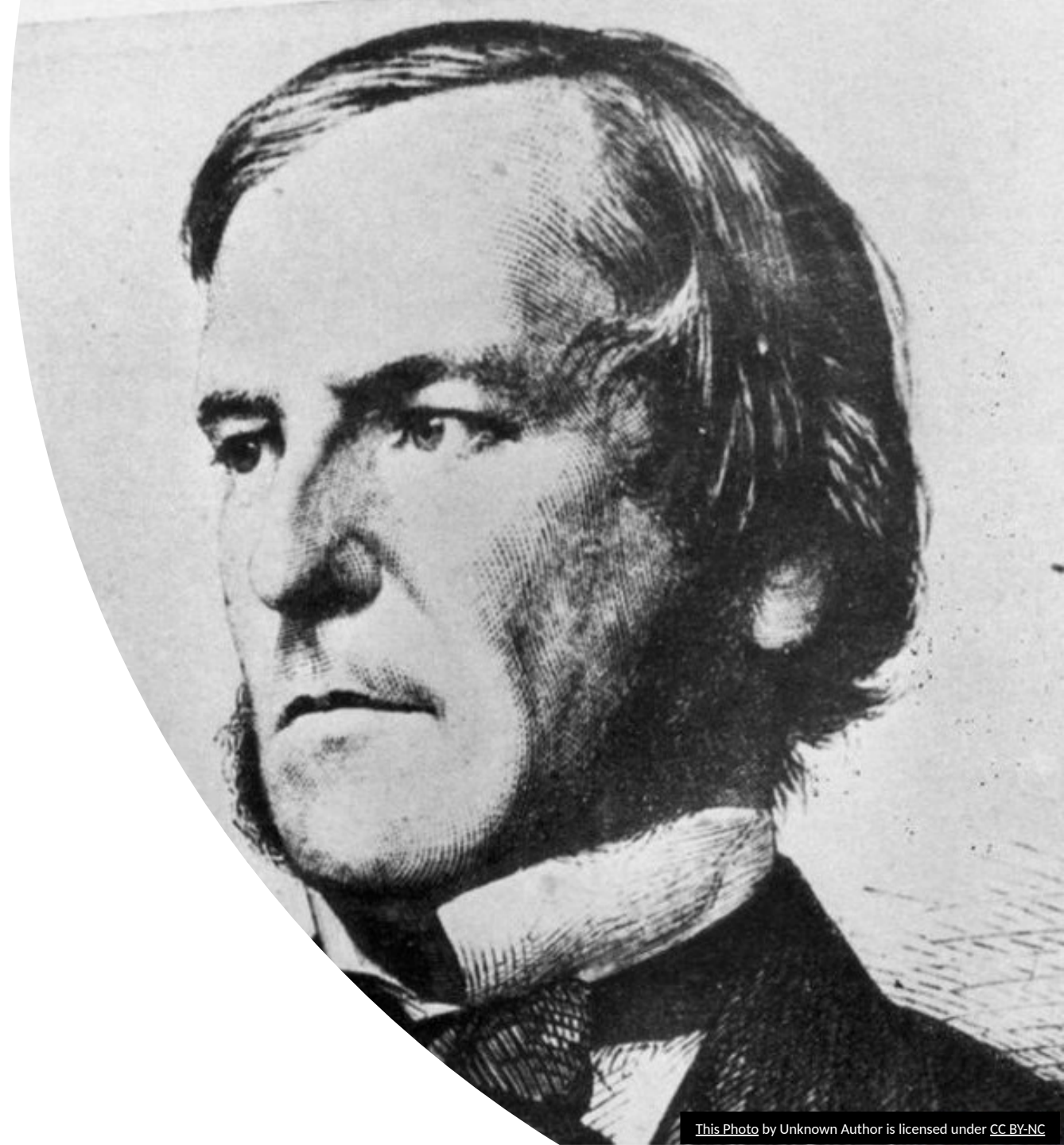
Needs to be a True or False: Boolean!



[This Photo](#) by Unknown Author is licensed under [CC BY-SA](#)

bool type in Python

- Short for **Boolean** ← Important term!
 - Basis of logic, named after George Boole, one of the great grandfathers of computing
- Has two values
 - True
 - False
 - Note the capital letter, different from most other languages!



Comparison operators

<

- Less than

>

- Greater than

==

- Equals

!=

- Doesn't equal

<=

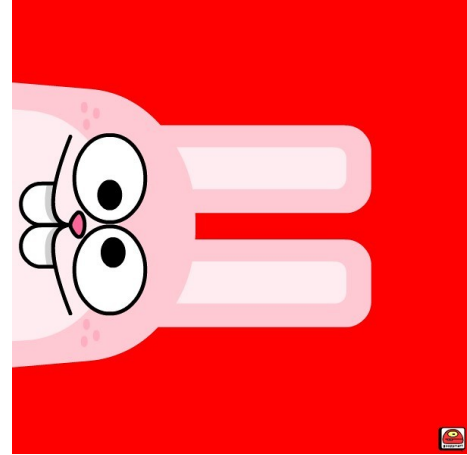
- Less than or equal to

>=

- Greater than or equal to

Single vs Double Equals

- In math, there is one meaning to the equals sign
 - Expresses the relationship that two things are equal
- In computer science, there are TWO equals signs
 - = used for variable assignment
 - `a = 6` puts the value 6 into the variable a
 - == used for value comparison
 - `a == 6` evaluates to True if the variable a has the value 6, False otherwise
 - **Very important to keep these straight when programming!**



Basic Boolean expressions: comparisons

- Demo
- $(6 < 7)$
 - Evaluates to True
- $(5 \geq 6)$
 - Evaluates to False
- $6 = 6$
 - No! Assignment!
- $6 == 6$
 - Evaluates to True
- $7 \neq 6$
 - Evaluates to True



[This Photo](#) by Unknown Author is licensed under [CC BY-NC-ND](#)

How do we use these things again?

- If statements!

if *expression*:

 #inside if

 #still inside if

#outside if

Note: no parentheses or braces!

Demo



if/else

if *expression*:

 #happens if expression is true

 #still inside if

else:

 #happens if expression is false

 #still inside if

#happens no matter what



if/elif/else

if *expression1*:

 #inside if

 #happens if exp1 True

elif *expression2*:

 #inside elif

 #happens if exp1 is False and

 # exp2 is True

else:

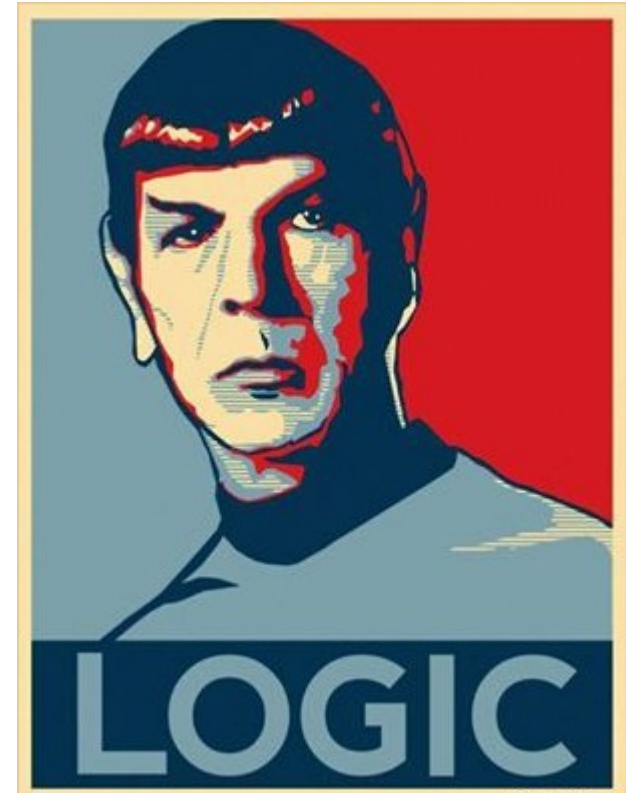
 #happens if both exp1 & exp2 are False



[This Photo](#) by Unknown Author is licensed under [CC BY-SA-NC](#)

More complicated logic...

- Sometimes we want to use logic (and, or, not) to combine things
- Python makes it pretty easy:
 - And is represented by and
 - Or is represented by or
- $(x > 4 \text{ or } y > 3)$
- demo



Not is a bit different

- X AND Y
- X OR Y
- NOT X
- AND and OR and binary operators (take two things)...
- NOT is unary
- In python, works like print() or the type converters
 - `not(x==4 OR x == 6)`



[This Photo](#) by Unknown Author is licensed under [CC BY-SA](#)

Saying number in range

- Very common in data science (need to check if number in proper range)
- In most programming languages, the only way to do it is like this:
 - If $10 < x$ and $x < 20$:
- Python supports chained comparisons!
- Can just say
 - if $10 < x < 20$:

Lab out

- if statements and comparisons